

Executive Summary

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Part 1: Executive Summary

We estimate that 401(k) plan eligibility increases net household financial assets by \$9,581 (95% CI: [\$6,682, \$12,481]).

Situation: US households face a widening retirement savings gap. Employer-sponsored 401(k) plans are the dominant accumulation vehicle for middle-income families, but coverage is uneven: roughly 60% of full-time private-sector workers have access, and part-time, seasonal, and gig workers are largely excluded. A financial services firm is evaluating whether extending 401(k) eligibility to its 50,000 part-time employees will meaningfully increase their retirement savings — or merely reshuffle existing wealth into a new account wrapper.

Complication: Naive comparisons show a \$19,559 gap in net financial assets between households with and without 401(k) access — but this is severely confounded by income selection. Workers at firms offering 401(k) plans earn \$15,368 more per year on average. Any simple comparison picks up the “high-quality employer” bundle, not the causal effect of eligibility itself.

Resolution: We apply Double Machine Learning (Chernozhukov et al., 2018) to 9,915 households from the Survey of Income and Program Participation (SIPP). DML uses gradient-boosted trees to nonparametrically partial out income, age, education, family structure, and savings behavior from both outcome and treatment, then estimates the residual causal relationship. The resulting ATE of \$9,581 is roughly half the naive figure — confirming substantial selection bias — but remains economically large and precisely estimated.

We recommend that the client extend 401(k) eligibility to currently ineligible workers. At 30% take-up among 50,000 newly eligible employees, this implies approximately \$143.7 million in new aggregate retirement savings — a compelling return on plan administration costs of roughly \$25 million per year.

Key assumption that could invalidate this: *Conditional independence — that after controlling for income, age, education, family size, marital status, IRA participation, and homeownership, which employer offers 401(k) access is effectively unrelated to unobserved savings preferences.*